

Tableau or Power BI Dashboarding project proposal

1. Executive Summary:

This project investigates whether Hollywood has shifted away from mid-budget filmmaking toward a “barbell” of micro-budget and tentpole releases — a widely repeated industry narrative that has not been rigorously tested against data. Using the TMDb 5000 Movie Dataset (4,803 films, 1990–2015), the project performs a full exploratory data analysis. The dashboards will focus on key performance indicators, providing a visual summarization of the findings for a non-technical audience.

2. Problem Statement:

Background: “Mid-budget movies are dead” is a widely repeated claim in entertainment journalism and industry commentary, but it is rarely backed by a transparent, reproducible analysis — most versions of the claim are anecdotal, citing a handful of high-profile examples rather than the full distribution of releases over time.

Objective: Quantify how the distribution of film budgets, box-office ROI, and genre composition has actually shifted between 1990 and 2015, and identify which parts of the common narrative hold up against the data and which don’t.

Scope: US/international theatrical releases with reported budget and revenue figures in the TMDb dataset, 1990–2015. Excludes direct-to-streaming and made-for-TV titles, which the dataset does not reliably capture.

3. Data Sources:

Primary Data: TMDb 5000 Movie Dataset (Kaggle/TMDb) — budget, revenue, genre, release date, runtime, and audience rating for 4,803 films.

Secondary Data: US Consumer Price Index (CPI) figures for inflation-adjusting budgets/revenue to constant 2015 dollars, and published US/Canada theatrical release-count statistics, used as an independent benchmark to test the dataset for sampling bias.

4. Methodology:

Data Integration: Load the raw CSV, resolve disguised missingness (budget/revenue fields zero-filled rather than left null), parse the stringified JSON genre field, parse release dates, and restrict the analysis window to 1990–2015 for adequate and consistent coverage.

Dashboard Design: Adjust all budgets/revenue to constant 2015 dollars, bucket films into four industry-standard budget tiers, and analyze tier share, ROI, and median budget over time using pandas and matplotlib — the same tier and trend breakdowns are then carried into the dashboard's visual layout.

Interactivity: Formally test the dataset's low-budget "surge" against an independent theatrical-release benchmark to rule out sampling bias, then fit an OLS log-log regression to quantify how budget, genre, runtime, and rating predict revenue; expose the tier filters and modeling results as clickable, drill-down elements in the dashboard.

5. Expected Outcomes:

- An interactive Tableau dashboard providing real-time exploration of budget-tier trends, ROI, and genre shifts across 25 years of film releases.
- A clear, evidence-based verdict on which parts of the "mid-budget is dead" narrative the data supports (the \$60–120M tier, not the \$20–60M tier, is what actually shrank) and which are overstated or artifacts of the data source.
- An interactive dashboard presenting the budget-tier timeline, the regulatory/ROI trends, and the bias-check findings for a non-technical audience.

6. Tools and Technologies:

- Tableau/Power BI for dashboard development.
- Python (pandas, NumPy) for data cleaning and feature engineering.
- Jupyter for the reproducible analysis notebook.

7. Risks and Challenges:

- The dataset is a popularity-ranked subset of TMDB's catalog, not a full census — catalog coverage is demonstrably uneven across years, requiring an explicit bias check (Section 4 of the analysis) before trusting any “more films of type X over time” finding.

- Budget and revenue figures are self/studio-reported and not independently audited; some are rounded industry estimates rather than verified figures.

- Marketing spend, a major driver of box-office performance, is not present in the dataset and is a significant omitted variable.

8. Conclusion:

This project is designed to move the “mid-budget movie is dead” conversation from anecdote to evidence. By combining a rigorous, reproducible cleaning and modeling pipeline with an explicit sampling-bias check and an accessible dashboard, it demonstrates not just what the data shows, but how confident we should be in each finding — a distinction most popular treatments of this topic skip entirely.